Computer Networking Guide: Architecture and Protocols.
Chapter 1: The OSI Reference Model.
The Open Systems Interconnection model defines seven architectural layers:
Layer 1 Physical Layer transmits raw bit streams over cables and radio frequencies.
Layer 2 Data Link Layer uses MAC hardware addresses to transfer frames across switches.
Layer 3 Network Layer is responsible for logical IP packet routing across routers.
Layer 4 Transport Layer provides host-to-host communication using TCP or UDP protocols.
Layer 5 Session Layer manages dialogues and connection sessions.
Layer 6 Presentation Layer handles syntax translation and SSL/TLS data encryption.
Layer 7 Application Layer supports end-user protocols including HTTP, DNS, and SMTP.

Chapter 2: The Domain Name System (DNS).
DNS translates alphanumeric domain names into numeric IP addresses.
DNS servers primarily listen on UDP port 53 for low-overhead query resolution.
However, when DNS responses exceed 512 bytes, DNS automatically falls back to TCP port 53.
Recursive resolvers query root name servers, Top-Level Domain (TLD) servers, and authoritative servers.
DNS caching reduces query latency across local networks and operating systems.

Chapter 3: Transmission Control Protocol (TCP) and UDP.
TCP provides reliable, ordered, and error-checked delivery of a stream of bytes.
To establish a connection, TCP performs a three-way handshake:
Step 1: The client sends a SYN (synchronize) packet to the server.
Step 2: The server responds with a SYN-ACK (synchronize-acknowledge) packet.
Step 3: The client replies with an ACK (acknowledge) packet.
TCP uses sequence numbers and sliding window flow control to prevent buffer overflows.
Conversely, UDP is a connectionless datagram protocol without handshakes or retransmission.